

Kei Hiroshima

hiroshima-kei-st@ynu.jp · keihiroshima.github.io · github.com/KeiHiroshima · linkedin.com/in/kei-hiroshima [Google Scholar](#) ·

Citizenship: Japan

Research interests

Continual Learning, Transfer Learning, Large Language Models, Model Merging, Black-Box Optimization, Natural Language Processing

Education

- | | |
|----------------|--|
| 2025 – Present | Yokohama National University – Yokohama, Japan
Ph.D. in Information Science, Graduate School of Environment and Information Sciences, Shirakawa/Uchida Laboratory (Supervisors: Prof. Shinichi Shirakawa, Asst. Prof. Ayato Uchida).
<i>Expected graduation: Spring 2029.</i> |
| 2023 | University of Technology Sydney (UTS) – Sydney, Australia
Exchange Student, Faculty of Engineering and Information Technology
<i>January – December 2023.</i> |
| 2022 – 2025 | Yokohama National University – Yokohama, Japan
M.S. in Informatics, Graduate School of Environment and Information Sciences |
| 2018 – 2022 | Yokohama National University – Yokohama, Japan
B.E. in Electrical and Computer Engineering, College of Engineering Science Department of Mathematics, Physics, Electrical Engineering and Computer Science |

Research experience

2022 – Present **Graduate Researcher, Shirakawa/Uchida Laboratory** (Yokohama National University)
Supervisors: Prof. Shinichi Shirakawa, Asst. Prof. Kento Uchida.

Publications

- 2026 **Tunable MAGMAX: Preference-Aware Model Merging for Continual Learning**
K. Hiroshima, K. Uchida, and S. Shirakawa.
Proceedings of the 28th International Conference on Pattern Recognition (ICPR 2026), Lyon, France. *To appear*.
- 2026 **BBOWP-Bench: Evaluating LLMs on Black-Box Optimization Word Problems**
Y. Yamada, K. Hiroshima, N. Yoshinari, K. Uchida, and S. Shirakawa.
Advances in Neural Information Processing Systems (NeurIPS 2026), Sydney, Australia.
Under review.
- 2026 **Autoformulation Benchmark Suite of Black-Box Optimization Problems Described in Natural Language**
Y. Yamada, K. Hiroshima, N. Yoshinari, K. Uchida, and S. Shirakawa.
The 40th Annual Conference of the Japanese Society for Artificial Intelligence, 2026, Gunma, Japan.
- 2025 **Enhancing Atherosclerosis Risk Prediction with Strategic Feature and Case Selections in Large Language Model**
H. Murase, K. Hiroshima, K. Uchida, M. Ohashi, N. Kashihara, A. J. Viera, H. Daida, S. Shirakawa, and Y. Yano.
ESC Congress 2025, Madrid, Spain.
- 2025 **Controlling Forgetting via Model Merging in Continual Learning**
K. Hiroshima, K. Uchida, and S. Shirakawa.
39th SSI 2025, Iwate, Japan.
[SSI Excellent Presentation Award]

Honors and awards

2025 **SSI Excellent Presentation Award**

SSI 2025 – SICE Symposium on Systems and Information, Iwate, Japan.

Awarded for the presentation “Controlling Forgetting via Model Merging in Continual Learning.”

Teaching experience

2025 – 2026 **Part-time Lecturer — Introduction to Mathematics, Data Science and AI**
Yokohama National University, Yokohama, Japan.

Technical skills

Programming languages

Proficient in: Python

Languages

Japanese (native), English (TOEIC 860 / IELTS 6.5: L 6.0 / R 7.0 / W 6.0 / S 6.0)